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Hybrid Advantage-Priority Hungarian Assignment MAPPO for Multi-UAV Cooperative Air Combat Decision-Making

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***Abstract*— This paper studies the multi-UAV cooperative air-combat problem and presents a hybrid Advantage-Priority Hungarian Assignment and Multi-Agent Proximal Policy Optimization (APH-MAPPO) framework for cooperative decision-making under dynamic target assignment and UAV attrition. First, the framework integrates attack-priority target allocation with residual Hungarian matching and continuous MAPPO maneuver control. The assignment is recomputed as combat geometry evolves, yielding target-conditioned friendly maneuver decisions. Second, it constructs and evaluates a reward composition that combines engagement advantage and shootdown outcome with allied proximity and shared-exposure penalties. A dual-flag masking mechanism is used as a fixed-dimensional implementation and training-interface design after UAV attrition. Extensive experiments in 3v3, 5v5, and asymmetric 3v5 confrontations show that APH-MAPPO achieved higher final mean reward and win rate compared with baseline methods including GA-MCTS, CA_MADDPG_AFF and LIORL.**

Manuscript received May 11, 2026; revised August 06, 2026; accepted September 08, 2026.
This work is supported by the National Natural Science Foundation of China (under grant 61903110), and Natural Science Foundation of Zhejiang Province (under grants LY24F030009 and LQ24F030006).

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***Index Terms*— multi-agent systems, multi-agent reinforcement learning, target assignment, autonomous maneuver decision-making**

I. Introduction

Recent years have witnessed the rapid advancement of manufacturing, artificial intelligence, and unmanned systems. As a representative category of unmanned systems, unmanned aerial vehicles (UAVs) have been extensively deployed in civilian and military missions. Meanwhile, their applications in modern air-combat scenarios have been continuously expanding. Air-combat decision-making empowered by artificial intelligence requires unmanned combat platforms to achieve target neutralization while ensuring self-preservation against adversarial threats [1], [2].

Early research on air-combat decision-making primarily relied on state-machine methods [3], [4], game-theoretic models [5]–[7], and traditional heuristic artificial-intelligence algorithms. Specifically, non-cooperative games or zero-sum matrix games are frequently incorporated into confrontations between multiple aircraft, aiming to allocate attack targets or plan optimal flight trajectories through solving Nash equilibrium.

However, these traditional methods can face limitations in complex and dynamically evolving air-combat scenarios [8], [9]. First, as the UAV swarm scale expands, the number of agents participating in the game increases sharply, leading to rapid growth of the joint strategy space [10]. Traditional mathematical optimization and game-theory algorithms can therefore become computationally demanding. Second, under information asymmetry and high dynamics, theoretical Nash equilibria based on idealized assumptions may be difficult to solve analytically in real time [5]. In addition, traditional game models rely heavily on artificially constructed payoff matrices based on domain expertise and prior experience. These considerations motivate adaptive decision methods for unknown battlefield situations and dynamically changing opponent strategies.

With the continuous improvement of computing power, reinforcement learning has been increasingly applied in the field of air-combat decision-making [11], [12]. Through continuous interaction with the environment [9], [13], aircraft agents can learn strategies such as long-range approach and close combat, thereby adapting to complex battlefield environments and diverse action-decision requirements [14], [15]. The simulated AlphaDogfight Trials further illustrated the potential of reinforcement learning in high-dimensional decision-making spaces [16].

With the aid of reinforcement learning, the core of multi-UAV cooperative combat often lies in solving a two-layer problem: high-level discrete target assignment and low-level continuous maneuver control. In terms of target assignment strategies, traditional frameworks usually rely on mathematical optimization or heuristic algorithms. For

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instance, recent studies have applied bio-inspired methods such as the attention-embedded lion swarm algorithm to solve dynamic task assignment problems [17], while others have combined improved genetic algorithms (GA) with Monte Carlo Tree Search (MCTS) auction protocols to construct dynamic assignment mechanisms in uncertain environments [18]. These approaches provide structured assignment mechanisms. In the highly dynamic air-combat setting considered here, however, the assigned target can become mismatched with the instantaneous maneuvering advantage of an aircraft. This motivates coupling target assignment more closely with maneuver decisions.

To bridge this maneuver-assignment gap, Wang et al. proposed a hierarchical virtual Bayesian reinforcement learning method [14], which effectively reduces the complexity of simulation computation. Xu et al. presented a reinforcement learning method incorporating virtual opponents and value attention decomposition, which decomposes the multi-UAV air-combat confrontation problem into multiple one-to-one subproblems for solution [19]. In contrast, the present three-degree-of-freedom (3-DOF) formulation focuses on explicitly coupling dynamic target assignment with target-conditioned continuous maneuver learning. In addition, advanced architectures such as Context-Aware Multi-Agent Deep Deterministic Policy Gradient (CA-MADDPG-AFF) [20] and Learned Implicit Opponent Strategy Reinforcement Learning (LIORL) [21] have been developed to infer adversarial intentions and optimize pursuit-evasion maneuvers [22], [23]. Efforts have also been made to leverage transfer learning to improve the scalability of MARL models under various swarm sizes [9]. More recently, Zheng et al. developed an asynchronous hierarchical framework with variable-length target-intention representations and a graph-based critic for distributed multi-UCAV air combat, while Li et al. studied independent soft actor–critic learning to improve multi-agent training stability and scalability [24], [25]. Together with the CA-MADDPG-AFF and LIORL methods discussed above, these studies improve situation representation, opponent modeling, hierarchical coordination, and policy optimization in multi-UAV air combat. Nevertheless, the task-level interface between target selection and continuous maneuver control remains insufficiently specified. Although assignment-assisted MARL has been explored [9], [22], a changing engagement may contain both immediately attackable pairs and remaining UAVs that still require allocation. How to preserve the feasible engagements, assign the residual pairs according to their current posture, and condition subsequent maneuver decisions on the resulting assignment remains an open design issue.

For the formulation considered in this study, three task-specific design requirements remain relevant. First, target assignment and maneuver control must be coupled as combat geometry evolves. Second, UAV attrition must be accommodated within a fixed-dimensional learning interface. Third, the reward should evaluate not only the engagement quality of each UAV against its assigned target, but also the interactions among friendly UAVs. Existing reward functions mainly emphasize local geometric advantages, firing or damage events, and terminal combat outcomes [9], [20], [22], [24]. They do not explicitly characterize the trade-off between individual engagement effectiveness and team-level risks, such as excessive proximity among friendly UAVs or their simultaneous exposure to the same adversarial UAV. How to incorporate these inter-UAV effects into the reward under dynamic target assignment therefore remains unclear.

Accordingly, this paper presents a hybrid Advantage-Priority Hungarian Assignment and Multi-Agent Proximal Policy Optimization (APH-MAPPO) framework for multi-UAV cooperative air combat. Its two task-level contributions are as follows:

(1) We integrate attack-priority target assignment with Hungarian matching into MAPPO-based motion control. This hierarchical framework reduces the combinatorial search space and enables efficient policy learning for target-oriented maneuver generation.

(2) We formulate a reward function that jointly considers attack and posture advantage, shootdown reward, allied proximity, and shared exposure. The proposed reward design promotes successful engagements while encouraging cooperative risk-aware behaviors and accelerating training convergence.

Under the specified simulation protocol, comparisons with selected baselines and component ablations are conducted in 3v3, 5v5, and asymmetric 3v5 scenarios. APH-MAPPO achieves higher final mean reward and win rate than the implemented baselines under the evaluated conditions, and the ablations quantify the effects of the assignment interface and reward composition, confirming that both components contribute to enhanced cooperative combat effectiveness.

The rest of the paper is structured as follows: Section II presents the aircraft dynamic model and several basic assumptions. Section III describes the APH-MAPPO design. Section IV reports simulation experiments and comparisons with three selected baseline methods. Finally, Section V concludes the paper.

II. Problem Formulation

A. Aircraft Dynamics Model

We use the 3-DOF point-mass model in Fig. 1, consistent with related air-combat studies [21], [26], to represent friendly and target UAVs. The motion equation of the i-th UAV can be expressed by Eq. (1):

$$\begin{cases} \dot{x}_i = v_i \cos\gamma_i \cos\psi_i \\ \dot{y}_i = v_i \cos\gamma_i \sin\psi_i \\ \dot{z}_i = v_i \sin\gamma_i \end{cases} \tag{1}$$

where $[x_i, y_i, z_i]^\top$ and v_i denote the position and scalar speed of the i-th UAV, respectively. For ease-of-notation, we introduce $\mathbf{p}_i = [x_i, y_i, z_i]^\top$ and $\mathbf{v}_i = \dot{\mathbf{p}}_i$ herein to stand

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for the position vector and velocity vector of the i-th UAV respectively. It is not hard to see that $|\mathbf{v}_i| = v_i$. Here, γ_i is the flight-path angle, defined as the angle between $\mathbf{v}_i$ and the horizontal oxy plane, and ψ_i is the heading angle, defined as the angle between the projection of $\mathbf{v}_i$ onto the oxy plane and the positive ox-axis. The dynamic model of the i-th UAV can be expressed by Eq. (2):

$$\begin{cases} \dot{v}_i = g(\zeta_i - \sin\gamma_i) \\ \dot{\gamma}_i = \dfrac{g}{v_i}(\hat{\zeta}_i \cos\mu_i - \cos\gamma_i) \\ \dot{\psi}_i = \dfrac{g\hat{\zeta}_i \sin\mu_i}{v_i \cos\gamma_i} \end{cases} \tag{2}$$

where μ_i is the bank angle, and ζ_i and $\hat{\zeta}_i$ are the dimensionless tangential and normal load factors, respectively. The corresponding acceleration components are $g\zeta_i$ and $g\hat{\zeta}_i$, where g is the gravitational acceleration. The flight state is controlled by the maneuver command $(\mu_i, \zeta_i, \hat{\zeta}_i)$.

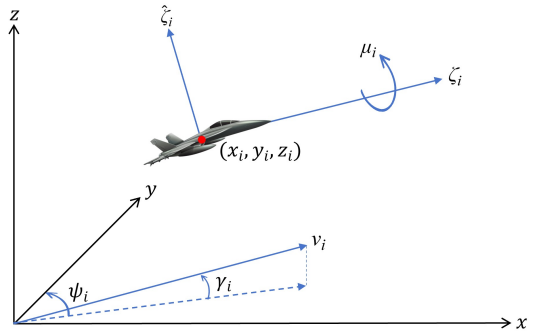


Fig. 1. three-degree-of-freedom particle model

B. Environment Description

As in [27], we adopt a forward-looking spherically capped cone as the attack-region assumption, whose apex is located at the UAV's position, and whose axis is aligned with the velocity vector. This implies that the onboard sensors are fixedly mounted and forward-looking, which is a relatively strong but practically relevant assumption for air-to-air engagement scenarios. As shown in Fig. 2, since we adopt a point-mass model in this paper, the attack region Ω of UAV i is defined as a forward-looking spherically capped cone with its apex at the position $\mathbf{p}_i$ and its central axis aligned with the instantaneous velocity vector $\mathbf{v}_i$. Its radial extent equals the effective missile range $\tau_{\max}$ and its angular half-aperture equals the maximum observation angle $\alpha_{\max}$.

The distance d_{ij} between the i-th UAV and the j-th UAV is defined as:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}. \tag{3}$$

Let α_i and α_j respectively denote the angle between $\mathbf{v}_i$ and $\mathbf{p}_j - \mathbf{p}_i$, and the angle between $\mathbf{v}_j$ and $\mathbf{p}_j - \mathbf{p}_i$. Let θ_{ij} represents the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$. During the entire air combat, we assume that each UAV has

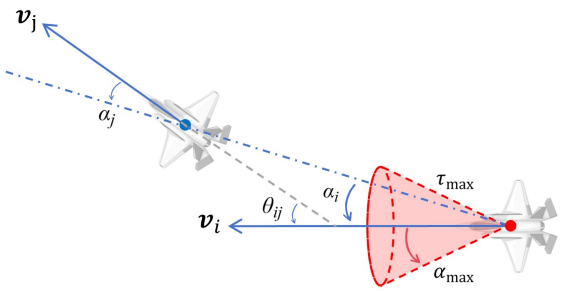


Fig. 2. Relative Geometry and Kinematic Variables

access to the velocities, relative distances, and the angles between the velocity and relative-position vectors of all other enemy UAVs (e.g., via omnidirectional radar), along with its own pose. It can also acquire such information for friendly UAVs, e.g., via communication. Admittedly, this assumption is somewhat conservative, yet it is commonly adopted in cooperative air-combat studies to maintain tractability [9], [22]

During air combat, a target UAV is regarded as neutralized and removed from the scenario once it remains within the attack region of any friendly UAV for a predefined dwell time. Furthermore, the friendly fleet is declared victorious if all adversarial UAVs are eliminated within the specified time horizon, at which point the simulation terminates; otherwise, the friendly fleet is considered defeated.

III. Our Method

In this section, a hybrid framework combining advantage-priority Hungarian assignment and the Multi-Agent Proximal Policy Optimization(APH-MAPPO) is presented to address the multi-UAV cooperative air combat problem. Heuristic assignment algorithms are employed to reduce the complexity of the multi-agent system, while the MAPPO algorithm is utilized to achieve adaptive and cooperative maneuver decision-making in dynamic air combat scenarios.The overall framework of the approach is illustrated in Fig. 4.

A. Target Assignment

The assignment is evaluated at every decision instant before friendly maneuver actions are issued. Pairwise attack and posture advantages are recomputed for all active friendly–adversarial pairs, and Algorithm 1 produces a candidate assignment matrix that determines the target-conditioned observation and reward for each friendly UAV. The previously locked target is retained when this matrix is unchanged; reassignment takes effect only when the newly computed assignment differs from that at the preceding decision instant. Following UAV attrition, destroyed UAVs are removed from the corresponding

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active set before this evaluation through the dual-flag mechanism (Section III-D). The present implementation does not use a separate switching threshold, hysteresis term, switching penalty, or ambush detector. The overall assignment process is illustrated in Fig. 3.

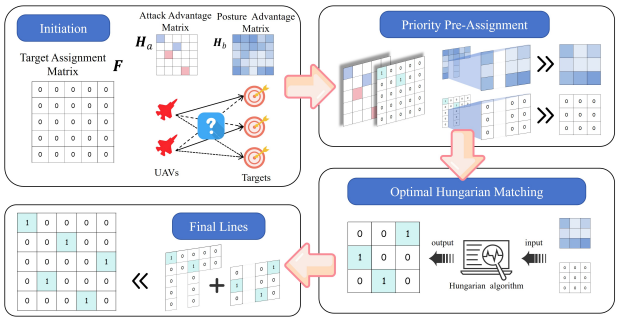


Fig. 3. Illustration of the Target Assignment Process

1. Attack Advantage Index

A target is regarded as shot down after it remains within the attack region Ω for a continuous time window. The attack advantage of friendly UAV i over adversarial target $\hat{j}$ is

$$\eta_{i,\hat{j}} = \begin{cases} \eta, & \text{if } (x_{\hat{j}}, y_{\hat{j}}, z_{\hat{j}}) \in \Omega_i \text{ and } -\frac{\pi}{2} < \alpha_{\hat{j}} < \frac{\pi}{2} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

The net attack advantage is the difference between the friendly and adversarial advantages:

$$\overline{r}_{i\hat{j}} = \eta_{i,\hat{j}} - \eta_{\hat{j},i}. \tag{5}$$

2. Posture Advantage Index

The posture advantage is defined in three distance regimes: below the safe distance, within attack range, and outside attack range:

$$\hat{r}_{i\hat{j}} = \begin{cases} -\dfrac{1}{d_{i\hat{j}}}, & d_{i\hat{j}} < \tau_{\min} \\ \cos\alpha_i + \cos\alpha_{\hat{j}}, & d_{i\hat{j}} \in [\tau_{\min}, \tau_{\max}] \\ \dfrac{v_i - v_{\min}}{v_{\max} - v_{\min}} \beta_{i\hat{j}}, & d_{i\hat{j}} > \tau_{\max} \end{cases} \tag{6}$$

where $\beta_{i\hat{j}} = \cos\alpha_i + \max\left(0, \cos\theta_{i\hat{j}}\right)$.

The additive form $\cos\alpha_i + \cos\alpha_{\hat{j}}$ distinguishes tail-chase direction: when UAV i tail-chases UAV $\hat{j}$, both terms equal 1 (score 2); when tail-chased, both equal -1 (score -2). Head-on geometries yield approximate cancellation. A product is not used because $(1)(1)$ and $(-1)(-1)$ are indistinguishable; this index is used only for posture-based residual target ranking after attack-range pre-assignment.

Based on the above analysis, three core matrices are defined to formalize the target assignment problem as follows:

1. **Attack advantage matrix** $\boldsymbol{H}_a \in \mathbb{R}^{N\times M}$:
   The element $\boldsymbol{H}_a(i,\hat{j})$ quantifies the attack advantage index of the i-th friendly agent against the $\hat{j}$-th adversarial target. Specifically, $\boldsymbol{H}_a(i,\hat{j}) = \eta$ indicates that the $\hat{j}$-th target UAV falls within the effective attack range of the i-th friendly UAV; Otherwise, the target is outside this range.
2. **Posture advantage matrix** $\boldsymbol{H}_b \in \mathbb{R}^{N\times M}$:
   The element $\boldsymbol{H}_b(i,\hat{j})$ characterizes the posture advantage index of the i-th friendly agent relative to the $\hat{j}$-th adversarial target. A higher value of this element signifies a more pronounced posture advantage for the friendly UAV in the engagement.
3. **Target assignment matrix** $\boldsymbol{F} \in \mathbb{R}^{N\times M}$:
   A binary indicator $\boldsymbol{F}(i,\hat{j}) = 1$ denotes that the $\hat{j}$-th adversarial target is assigned to the i-th friendly UAV; conversely, $\boldsymbol{F}(i,\hat{j}) = 0$ means the $\hat{j}$-th target is not allocated to the i-th friendly UAV.

When multiple targets simultaneously enter the effective attack range, each friendly UAV can lock onto at most one target. Accordingly, the priority pre-assignment must satisfy

$$\sum_{\hat{j}=1}^{M} \boldsymbol{F}(i,\hat{j}) \le L, \quad \forall i \in \{1,\dots,N\}, \tag{7}$$

where L denotes the maximum number of targets that a friendly UAV can lock onto simultaneously and is set to 1 in this work. This is a per-friendly-UAV single-target constraint rather than a global one-to-one assignment constraint: no constraint of the form $\sum_i \boldsymbol{F}(i,\hat{j}) \le 1$ is imposed on an adversarial target. Hence, several friendly UAVs may be assigned to the same target when the task-specific rules described below permit target reuse.

During the initial assignment, targets within the effective attack range are prioritized under a lexicographic rule because they offer immediately feasible engagement opportunities. If multiple eligible targets are provisionally assigned to the same friendly UAV, only the target with the highest posture advantage is retained to satisfy the per-UAV constraint $L = 1$. By contrast, if several friendly UAVs independently have immediate attack eligibility for the same target, their assignments are retained; this attack-priority target reuse is intentional because each retained assignment represents an immediately feasible engagement. The remaining UAV–target assignments are then determined by maximizing the total posture advantage:

$$\max \sum_{i=1}^{N}\sum_{\hat{j}=1}^{M} \boldsymbol{F}(i,\hat{j})\boldsymbol{H}_b(i,\hat{j}). \tag{8}$$

In order to adapt to the minimum-cost matching property of the Hungarian algorithm, the matrix is negated here. We define the cost matrix $\hat{\boldsymbol{H}}_b = -\boldsymbol{H}_b$, converting the maximum advantage into the minimum cost. In sum-

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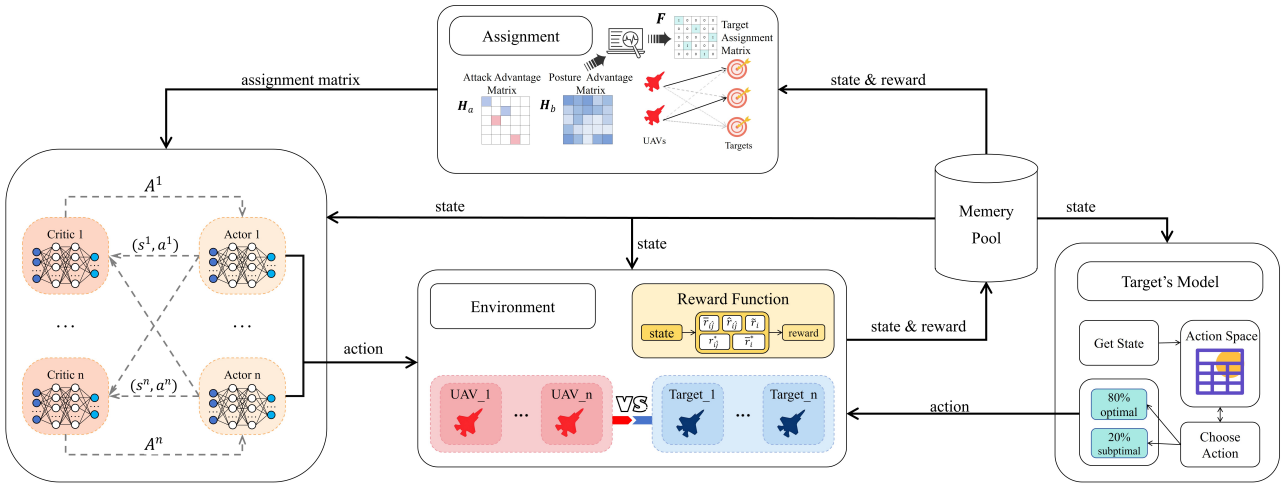


Fig. 4. Illustration of our method

mary, the optimization function is transformed into:

$$\begin{aligned} &\min \sum_{i=1}^{N} \sum_{\hat{j}=1}^{M} \boldsymbol{F}(i,\hat{j}) \cdot \hat{\boldsymbol{H}}_b(i,\hat{j}) \\ &\text{s.t.} \sum_{\hat{j}=1}^{M} \boldsymbol{F}(i,\hat{j}) \leq L, \quad \forall i \in \{1,\ldots,N\}. \end{aligned} \tag{9}$$

After priority pre-assignment, the rows and columns associated with fixed assignments are removed from $\hat{\boldsymbol{H}}_b$, yielding the residual cost matrix $\boldsymbol{C}_u \in \mathbb{R}^{n_u \times m_u}$. Here, n_u and m_u denote the numbers of unassigned friendly UAVs and residual targets, respectively. For $m_u > 0$, the matching matrix is constructed as

$$\boldsymbol{C}_{\text{match}} = \begin{cases} \boldsymbol{C}_u, & n_u \leq m_u, \\ \text{tile}(\boldsymbol{C}_u, (1, \lceil n_u/m_u \rceil)), & n_u > m_u. \end{cases} \tag{10}$$

The Hungarian algorithm is then applied to $\boldsymbol{C}_{\text{match}}$. Each selected column is mapped to its original target by modulo indexing. Consequently, each residual friendly UAV is assigned one valid target whenever $m_u > 0$. Within this residual phase, distinct target columns yield a one-to-one matching when $n_u \leq m_u$, whereas replication-induced target reuse occurs only when $n_u > m_u$. This residual-phase condition is separate from the attack-priority pre-assignment, in which multiple friendly UAVs may already retain the same immediately attackable target. The overall pseudocode is shown in Algorithm 1.

Unlike dummy-column padding, which represents an unmatched UAV through an artificial cost, every replicated column represents a valid adversarial target. This task-specific construction keeps each residual friendly UAV focused on an active target when $n_u > m_u$. Together with the attack-priority rule, it permits several friendly UAVs to engage the same target but does not impose an explicit per-target capacity constraint or a penalty for excessive target concentration; dummy-assignment and capacity-constrained formulations remain alternatives for future study.

B. Multi-Agent Proximal Policy Optimization

1. *Action Space*

We use a continuous action space based on the UAV dynamics model in Eq. (2). This representation provides continuously valued maneuver commands without discretizing each control variable. Equation (11) defines the action space of the i-th UAV:

$$a_i = [\mu_i, \zeta_i, \hat{\zeta}_i]^\top. \tag{11}$$

2. *State Space*

In the air combat environment, the state space of each UAV is defined as the combination of the observation space of the UAV agent itself, the target UAV and the friendly UAV. In this paper, the Eq. (12) below defined the observation space of the i-th intelligent UAV agent for the $\hat{j}$-th target drone:

$$s_{i\hat{j}} = [v_i, \gamma_i, \psi_i, \alpha_i, z_i, v_{\hat{j}}, \gamma_{\hat{j}}, \psi_{\hat{j}}, \alpha_{\hat{j}}, z_{\hat{j}}, d_{i\hat{j}}]. \tag{12}$$

Similarly, the observation space of the i-th UAV for the k-th friendly UAV is:

$$s_{ik} = [v_i, \gamma_i, \psi_i, \alpha_i, z_i, v_k, \gamma_k, \psi_k, \alpha_k, z_k, d_{ik}]. \tag{13}$$

where $i \neq k$. In summary, the state space $\hat{s}_i$ of the i-th UAV can be expressed as:

$$\hat{s}_i = \left[\bigcup_{\hat{j}=1}^{M} s_{i\hat{j}}, \bigcup_{k=1}^{N} s_{ik} \mid (k \neq i) \right]. \tag{14}$$

3. *Reward Function*

The maneuver policy uses a structured geometry-based reward with five components that encode engagement geometry, target neutralization, episode outcome, and allied proximity and shared exposure.

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Algorithm 1 Target assignment with attack-priority pre-assignment and residual Hungarian matching.

Input: Attack advantage matrix $\boldsymbol{H}_a$, posture advantage matrix $\boldsymbol{H}_b$, and per-UAV target limit $L = 1$.
Output: Target assignment matrix $\boldsymbol{F} \in \{0,1\}^{n\times m}$.
Let n and m denote the numbers of active friendly UAVs and adversarial targets, respectively.
Initialize $\boldsymbol{F} \leftarrow \mathbf{0}_{n\times m}$ and set $\boldsymbol{C} \leftarrow -\boldsymbol{H}_b$.
Phase 1: Attack-Priority Pre-assignment
for $i = 1$ **to** n **do**
  for $j = 1$ **to** m **do**
    if $\boldsymbol{H}_a[i,j] = \eta$ **then**
      $\boldsymbol{F}[i,j] \leftarrow 1$ {Provisionally assign an eligible target}
    end if
  end for
end for
Phase 2: Per-UAV Conflict Resolution
for $i = 1$ **to** n **do**
  while $\sum_{j=1}^{m} \boldsymbol{F}[i,j] > L$ **do**
    $j^- \leftarrow \arg\max_{j:\boldsymbol{F}[i,j]=1} \boldsymbol{C}[i,j]$ {Lowest posture advantage}
    $\boldsymbol{F}[i,j^-] \leftarrow 0$
  end while
end for
Phase 3: Residual Problem Construction
$I_a \leftarrow \{i \mid \exists j,\ \boldsymbol{F}[i,j] = 1\}$
$J_a \leftarrow \{j \mid \exists i,\ \boldsymbol{F}[i,j] = 1\}$
$I_u \leftarrow \{1,\dots,n\} \setminus I_a$
$J_u \leftarrow \{1,\dots,m\} \setminus J_a$
if $I_u = \emptyset$ **then**
  return $\boldsymbol{F}$
end if
if $J_u = \emptyset$ **then**
  return $\boldsymbol{F}$
end if
$n_u \leftarrow |I_u|$, $m_u \leftarrow |J_u|$
$\boldsymbol{C}_u \leftarrow \boldsymbol{C}[I_u, J_u]$
Phase 4: Residual Hungarian Matching
if $n_u \le m_u$ **then**
  $\boldsymbol{C}_{\text{match}} \leftarrow \boldsymbol{C}_u$
else
  $q \leftarrow \lceil n_u/m_u \rceil$
  $\boldsymbol{C}_{\text{match}} \leftarrow \text{tile}(\boldsymbol{C}_u, (1,q))$ {Replicate target columns}
end if
$(\boldsymbol{r}, \boldsymbol{c}) \leftarrow \text{HUNGARIAN}(\boldsymbol{C}_{\text{match}})$
for all $(r_k, c_k) \in (\boldsymbol{r}, \boldsymbol{c})$ **do**
  $i \leftarrow I_u[r_k]$
  $j \leftarrow J_u\big[1 + \big((c_k - 1) \bmod m_u\big)\big]$
  $\boldsymbol{F}[i,j] \leftarrow 1$
end for
return $\boldsymbol{F}$

a: Attack Advantage Reward

We first design the attack advantage reward, which follows the attack advantage index defined in Eq. (5) for the target assignment module.

b: Posture Advantage Reward

We further propose the posture advantage reward, which adopts a hierarchical formulation consistent with the posture advantage index defined in Eq. (6) in the target assignment module.

c: Allied Proximity and Shared-Exposure Penalty

In addition to the target-oriented rewards, we introduce an allied proximity and shared-exposure penalty to reduce concentrated maneuvers and shared adversarial exposure.

Let $k \neq i$ denote another friendly UAV and let $\hat{j}$ denote an adversarial UAV. We define the binary threat indicator

$$q_{\hat{j}\to i} = \mathbb{I}\left(\eta_{\hat{j},i} > 0\right), \tag{15}$$

where $q_{\hat{j}\to i} = 1$ indicates that friendly UAV i satisfies the attack conditions of adversarial UAV $\hat{j}$ under the same geometric criterion used in the attack advantage index.

For each pair of friendly UAVs i and k, define the shared-exposure indicator as

$$c_{ik} = \max_{\hat{j}} \left(q_{\hat{j}\to i} \cdot q_{\hat{j}\to k}\right), \tag{16}$$

where $c_{ik} = 1$ indicates that at least one adversarial UAV can simultaneously engage both friendly UAVs i and k.

The allied penalty for UAV i is defined as

$$\tilde{r}_i = -\frac{1}{N-1} \sum_{\substack{k=1 \\ k\neq i}}^{N} \left[\max\left(0, 1 - \frac{d_{ik}}{\tau_{\min}}\right) + c_{ik}\right]. \tag{17}$$

The penalty is evaluated at each environment step using the relative states of active UAVs in the simulator, where $\tau_{\min} > 0$ denotes the minimum safe separation. The first term penalizes overly close friendly UAVs, while $c_{ik} = 1$ indicates that at least one adversarial UAV can simultaneously engage both UAVs i and k. This penalty is used only in the MAPPO reward to encourage spatial dispersion and reduce shared exposure; it does not modify the Hungarian target assignment.

d: Shootdown Reward

In this paper, we design both an individual shootdown reward and a global shootdown reward as supplements to the above three immediate rewards. If the $\hat{j}$-th target UAV is continuously locked by the i-th UAV beyond the aforementioned threshold, the i-th UAV that performed the locking action obtains the individual shootdown reward, as shown in the Eq. (18) below:

$$r^*_{i\hat{j}} = \begin{cases} r_0, & t_{i\hat{j}} > \bar{t} \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

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e: Episodic Reward

If all target UAVs are shot down by our agents, all friendly UAVs obtain the episodic reward, as shown in the Eq. (19) below:

$$\overline{r}_i^* = \begin{cases} \hat{N} \cdot e_1 - \left\lfloor \dfrac{T}{e_2} \right\rfloor, & \hat{M} = 0 \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

where $\hat{N}$ denotes the number of our surviving UAV agents, $\hat{M}$ denotes the number of surviving enemy UAV targets, T denotes the total number of time steps from environment initialization to shooting down all enemy UAVs, and $\lfloor \cdot \rfloor$ denotes the floor function. e_1 and e_2 are coefficients greater than 1. It can be seen that the reward value is linked to the number of surviving UAVs and the number of combat steps : the more surviving UAVs and the fewer combat steps, the higher the reward.

Combined with the formulations above (Eq. (5), Eq. (6), Eq. (17), Eq. (18), and Eq. (19)), at each time step in the air combat model, the reward function Eq. (20) of the i-th UAV agent is computed as follows:

$$r_i = \overline{r}_{i\hat{j}} + \hat{r}_{i\hat{j}} + \tilde{r}_i + r^*_{i\hat{j}} + \overline{r}^*_i. \tag{20}$$

C. Target Aircraft Decision Model

For the rule-based adversarial policy used in this study, we use a 15-action discrete maneuver structure adapted from Zheng and Duan [28]. It combines five maneuver directions, namely forward flight, left turn, right turn, climb, and dive, with three speed modes. The parameter values in Tab I are specified for the present rule-based adversarial model: constant speed, acceleration, and deceleration. At each decision step, the adversarial UAV $\hat{j}$ selects one action from this library according to the prescribed rule-based policy, as listed in Tab I. In contrast, the friendly MAPPO policy directly outputs continuous maneuver commands.

TABLE I
Discrete maneuver set for an adversarial UAV $\hat{j}$

No.	Maneuvering Action	μ	$\hat{\zeta}$	ζ
1	Steady Forward	0	1	0
2	Accelerated Forward	0	1	2
3	Decelerated Forward	0	0	-1
4	Steady Left Turn	$-\pi/3$	8	0
5	Accelerated Left Turn	$-\pi/3$	8	2
6	Decelerated Left Turn	$-\pi/3$	8	-1
7	Steady Right Turn	$\pi/3$	8	0
8	Accelerated Right Turn	$\pi/3$	8	2
9	Decelerated Right Turn	$\pi/3$	8	-1
10	Steady Climb	0	8	0
11	Accelerated Climb	0	8	2
12	Decelerated Climb	0	8	-1
13	Steady Dive	π	8	0
14	Accelerated Dive	π	8	2
15	Decelerated Dive	π	8	-1

For the adversarial team, we use a fixed stochastic greedy policy. At each step, every active adversarial UAV evaluates the 15 maneuvers in Tab I. Candidate maneuver a produces the one-step virtual state $\tilde{\boldsymbol{x}}^{(a)}_{\hat{j},t} = f_b(\boldsymbol{x}_{\hat{j},t}, a)$. For each active friendly UAV $i \in \mathcal{F}_t$, its tactical score is

$$q_{\hat{j}i}(a) = \frac{\tilde{v}^{(a)}_{\hat{j},t} - v_{\min}}{v_{\max} - v_{\min}} \cos\left(\tilde{\alpha}^{(a)}_{\hat{j}}\right) + \tilde{\overline{r}}^{(a)}_{\hat{j}i}, \tag{21}$$

where the first term is the normalized enemy speed–aspect score and $\tilde{\overline{r}}^{(a)}_{\hat{j}i} = \tilde{\eta}^{(a)}_{\hat{j},i} - \tilde{\eta}^{(a)}_{i,\hat{j}}$ is the net attack advantage from Eq. (5), with the enemy UAV listed first. The distance term called by the implementation is identically zero and is therefore omitted. The action score is the sum over active friendly UAVs:

$$Q_{\text{sum}}(a) = \sum_{i \in \mathcal{F}_t} q_{\hat{j}i}(a). \tag{22}$$

Let $a_{[1]}$ and $a_{[2]}$ be the highest- and second-highest-ranked maneuvers, respectively. The executed action is

$$a_{\hat{j},t} = \begin{cases} a_{[1]}, & u < p, \\ a_{[2]}, & u \geq p, \end{cases} \qquad u \sim \mathcal{U}(0,1), \tag{23}$$

where p is a preset probability value ranging from 0 to 1. The value closer to 1 means a higher probability of selecting the top-ranked action, while the value closer to 0 means a higher probability of choosing the second-ranked action.

Two complementary post-training sensitivity tests vary one opponent-design factor at a time. The first changes the best-action probability while retaining Eq. (22). The second fixes the best-action probability at 0.8 and replaces the aggregation rule with

$$Q_{\text{opp}}(a) = \max_{i \in \mathcal{F}_t} q_{\hat{j}i}(a), \qquad Q_{\text{threat}}(a) = \min_{i \in \mathcal{F}_t} q_{\hat{j}i}(a). \tag{24}$$

The opportunistic-max variant favors a maneuver with a high score against at least one friendly UAV, whereas the threat-aware maximin variant ranks maneuvers by their least favorable pairwise score before applying the same stochastic action-selection rule. The adversarial policy operates independently of the friendly MAPPO policy and reacts to current combat geometry but does not learn within or across episodes. These tests therefore provide controlled evidence under several hand-designed, state-reactive rules; they do not establish robustness against independently learned or self-play opponents.

D. Dual-Flag Mechanism

In air combat, UAV attrition or changes in the initial number of friendly aircraft may alter the observation space, causing missing or mismatched inputs and limiting cooperative decision-making. To address this, we introduce a dual-flag mechanism to fix the observation and state spaces. Based on the observation space in Eq. (12), a supplementary flag $O_{i\hat{j}}$ is added, and the revised observation space of the i-th agent is:

$$s_{i\hat{j}} = [O_{i\hat{j}}, v_i, \gamma_i, \psi_i, \alpha_i, z_i, v_{\hat{j}}, \gamma_{\hat{j}}, \psi_{\hat{j}}, \alpha_{\hat{j}}, z_{\hat{j}}, d_{i\hat{j}}] \tag{25}$$

where $O_{i\hat{j}}$ acts as a flag that takes a value of either 0 or 1. Specifically, Let $\hat{O}^t_{i,p} \in \{0,1\}$ denote the validity

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of friendly UAV i in parallel environment p at decision step t. If $\hat{O}^t_{i,p} = 0$, the UAV is excluded from flight-state propagation, target assignment, and allied-safety evaluation. Its observation slot is set to zero so that the observation and action tensor dimensions remain fixed.

For each friendly UAV i, the current-state mask is stored with the transition and is used to reduce the actor and critic losses only over valid samples:

$$\mathcal{L}_{\pi,i} = \frac{\sum_{(t,p)\in\mathcal{B}} \hat{O}^t_{i,p}\ell^t_{\pi,i,p}}{\sum_{(t,p)\in\mathcal{B}} \hat{O}^t_{i,p}}, \qquad \mathcal{L}_{V,i} = \frac{\sum_{(t,p)\in\mathcal{B}} \hat{O}^t_{i,p}\left(V_i(s^t_p) - R^t_{i,p}\right)^2}{\sum_{(t,p)\in\mathcal{B}} \hat{O}^t_{i,p}}, \tag{26}$$

where $\mathcal{B}$ denotes a mini-batch, $\ell^t_{\pi,i,p}$ is the entropy-regularized PPO surrogate loss, and $R^t_{i,p}$ is the return. If a mini-batch contains no valid sample for UAV i, its actor and critic updates are skipped.

The next-state mask prevents generalized advantage estimation from bootstrapping through an attrition state:

$$\delta^t_{i,p} = \hat{O}^t_{i,p}\left[r^t_{i,p} + \gamma\hat{O}^{t+1}_{i,p}V_i(s^{t+1}_p) - V_i(s^t_p)\right], \qquad A^t_{i,p} = \delta^t_{i,p} + \gamma\lambda b^t_p\hat{O}^{t+1}_{i,p}A^{t+1}_{i,p}, \tag{27}$$

where $b^t_p = 1$ for a nonterminal transition and $b^t_p = 0$ when the episode terminates. Thus, a UAV destroyed during step t retains the reward of that transition, but its invalid next state does not contribute to bootstrapping or subsequent gradient updates.

To preserve a fixed-size batch, the actor network may still produce a candidate action for an invalid slot. This incurs residual forward-pass computation; however, the action is neither applied to the simulator nor included in target assignment, actor loss, critic loss, or GAE. This implementation supports scenarios with up to five friendly UAVs by zero-padding unused observation slots. In this study, the 3v3, 5v5, and 3v5 configurations are trained and evaluated separately. Therefore, we do not claim zero-shot transfer of a policy trained at one team size to another team size.

IV. Simulation Results and Analysis

A. Experimental Setup

1. *Engagement Scenarios and Simulation Parameters*

The engagement environment contains N friendly UAVs and M adversarial UAVs in an unbounded three-dimensional space. The 3-DOF simulator uses a discrete time step of $0.25\,\mathrm{s}$, and every episode is limited to 250 decision steps. The shootdown criterion follows the two-step dwell-time rule in Section II, corresponding to $0.5\,\mathrm{s}$. The principal physical constraints are $\tau_{\max} = 2000\,\mathrm{m}$, $\tau_{\min} = 100\,\mathrm{m}$, $\alpha_{\max} = \pi/4$, $v_{\max} = 400\,\mathrm{m/s}$, and $v_{\min} = 100\,\mathrm{m/s}$.

The baseline reward coefficients are $(\eta, e_1, e_2) = (5, 20, 25)$. The attack-advantage weight η gives priority to feasible offensive opportunities, whereas the terminal coefficients make mission completion dominant over the accumulated time penalty within a 250-step episode. The continuous maneuver variables and their admissible ranges are listed in Tab II.

TABLE II
Continuous action-variable constraints.

Variable	Notation	Range
Bank angle	μ_i	$[-\pi, \pi]$
Tangential load factor	ζ_i	$[-1, 2]$
Normal load factor	$\hat{\zeta}_i$	$[0, 8]$

2. *Implementation and Training Protocol*

All policies were implemented in PyTorch. The network architecture and training hyperparameters are summarized in Tab III. The 3v3, 5v5, and 3v5 policies were trained separately, with zero padding used only to maintain a fixed network interface.

TABLE III
MAPPO implementation and training hyperparameters.

Item	Value
Actor hidden layers	$128, 128$ with $\tanh$
Critic hidden layers	$128, 128$ with $\tanh$
Actor output distribution	Beta distribution
Local observation dimension	12
Centralized critic input dimension	60
Continuous action dimension	3
Optimizer	Adam
Learning rate	3×10^{-4}
Discount factor γ	0.954
GAE coefficient λ	0.95
PPO clipping coefficient ϵ	0.2
Entropy coefficient	10^{-3}
PPO epochs per update	10
Mini-batch size	128
Parallel environments	10
Rollout horizon per environment	1500 steps
Maximum episode length	250 steps
Checkpoint interval	50 episodes

Unless otherwise stated, each configuration was trained for 5000 episodes using five independent seeds. For seed s, the ten training workers used seeds $1000s + r$, where $r \in \{0, \ldots, 9\}$.

Experiments were conducted on an AutoDL instance with 12 vCPUs of an Intel Xeon Platinum 8255C CPU, 43 GB memory, an NVIDIA GeForce RTX 2080 Ti GPU (11 GB), Ubuntu 22.04, PyTorch 2.1.2, and CUDA 11.8. The approximate training times per seed were 6, 8, and 10 hours for the 3v3, 3v5, and 5v5 scenarios, respectively.

3. *Baselines and Evaluation Metrics*

The comparison includes three baselines: GA-MCTS [18], CA_MADDPG_AFF [20], and LIORL [21]. GA-MCTS is treated as an online planning baseline rather than as a policy trained under the PPO episode budget. For trainable baselines implemented in the present environment, the common training configuration in Tab III is used unless their published method requires a method-specific

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setting. Because the original LIORL offline pretraining dataset was unavailable, its offline pretraining stage was omitted; the reward-weight ratios and attack-trigger conditions were retained in the present environment, and all other hyper-parameters strictly follow the original literature without modification.

The primary metric is the percentage of evaluation episodes in which all adversarial UAVs are eliminated. The ablation study additionally reports the number of surviving friendly UAVs and the cumulative proximity and shared-exposure risks defined by the safety terms in Eq. (17). Win rate and survivors are higher-is-better, whereas both risk metrics are lower-is-better. Two post-training opponent-policy sensitivity studies are reported separately from the primary multi-seed training comparison: one varies the best-action probability for a single frozen checkpoint, and the other varies the score-aggregation rule across five frozen 5v5 checkpoints. In the latter study, each checkpoint–opponent-mode combination is evaluated over 200 episodes, comprising 40 episodes for each of five evaluation seeds.

B. Overall Performance

1. *Multi-Seed Performance Across Team Configurations*

Fig 5 summarizes the final evaluations over five independently trained policies per team configuration. Each final checkpoint was evaluated over 200 episodes.

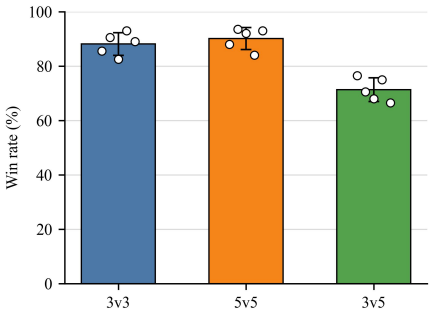


Fig. 5. Final win rate across five training seeds. Error bars denote one standard deviation across seeds.

APH-MAPPO achieved win rates of $88.10\% \pm 4.14\%$, $90.10\% \pm 4.04\%$, and $71.30\% \pm 4.34\%$ in the 3v3, 5v5, and 3v5 scenarios, respectively. The reported standard deviations quantify the observed variation across the five training seeds under the evaluated initialization and opponent settings. The lower 3v5 win rate reflects the greater difficulty of the asymmetric engagement.

2. *Action-Selection-Stochasticity Sensitivity in 5v5*

To examine the effect of action-selection stochasticity, we evaluated one fixed 5v5 checkpoint trained with the baseline best-action probability of 0.8 against three best-action probabilities while retaining the aggregate-sum score formulation in Eq. (22). For each probability, five evaluation seeds (100–104) and 200 episodes per seed were used. The mean win rates (sample standard deviation

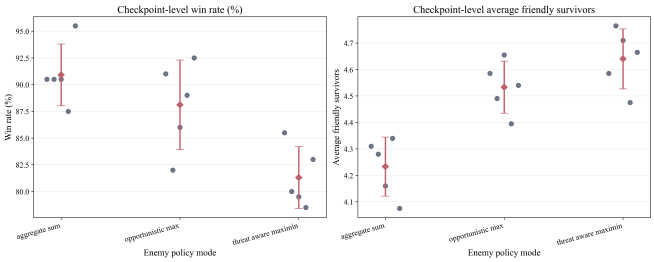


Fig. 6. Sensitivity of five 5v5 APH-MAPPO checkpoints to three opponent score-aggregation modes.

across evaluation seeds) were $94.3\% \pm 1.48\%$ for probability 1.0, $92.3\% \pm 2.49\%$ for the baseline probability 0.8, and $92.7\% \pm 0.45\%$ for probability 0.6. Thus, the results remained within a 2.0-percentage-point range across the tested settings. This single-checkpoint check is descriptive rather than evidence of an optimal opponent ratio or robustness against learned opponents.

3. *Opponent Score-Aggregation Sensitivity in 5v5*

We next examined a distinct source of opponent-policy variation by changing how each adversarial UAV aggregates its pairwise scores over the active friendly team. Unlike the preceding test, the best-action probability was fixed at 0.8 in all conditions. Five frozen 5v5 checkpoints, one from each independently trained seed at episode 4950, were evaluated against the aggregate-sum, opportunistic-max, and threat-aware maximin modes. Each checkpoint–mode combination was evaluated over 200 episodes, comprising 40 episodes for each of five evaluation seeds (100–104), yielding 3000 evaluation episodes in total.

As shown in Fig. 6, the aggregate-sum opponent produced a mean win rate of $90.9\% \pm 2.88\%$ and 4.233 ± 0.112 surviving friendly UAVs. Under the opportunistic-max opponent, the corresponding values were $88.1\% \pm 4.19\%$ and 4.533 ± 0.098, whereas the threat-aware maximin opponent produced $81.3\% \pm 2.89\%$ and 4.640 ± 0.113. All values are mean $\pm$ sample standard deviation across the five independently trained checkpoints. Relative to aggregate-sum within the same checkpoints, the mean win-rate changes were -2.8 and -9.6 percentage points for opportunistic-max and threat-aware maximin, respectively; the corresponding survivor changes were $+0.300$ and $+0.407$. Thus, the observed performance was not invariant to the opponent decision rule: the two alternative aggregations were associated with lower complete-elimination rates but higher numbers of surviving friendly UAVs. The scope of this hand-designed, non-learning comparison is stated in Section IV.

4. *Comparison with Baseline Methods*

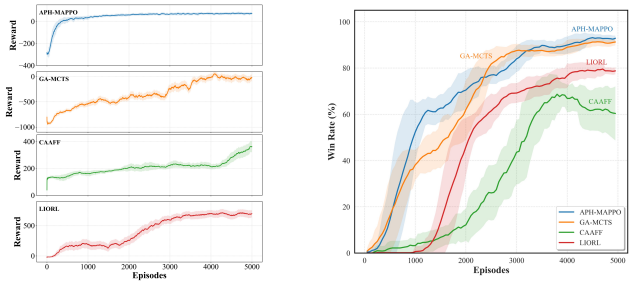
The baseline comparison uses the common simulation constraints and evaluation metrics defined above.

Fig 7 compares APH-MAPPO with the baseline methods in the 3v3 scenario. Each curve reports the mean

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over five independent random seeds, and the shaded region denotes one standard deviation across seeds. APH-MAPPO achieved the highest final mean reward and win rate among the compared implementations. Its learning curve also exhibited lower observed across-seed variability, as indicated by the narrower standard-deviation band. Thus, under the common 3v3 simulation and evaluation protocol, APH-MAPPO produced higher final mean performance with lower observed variability than the selected implementations.

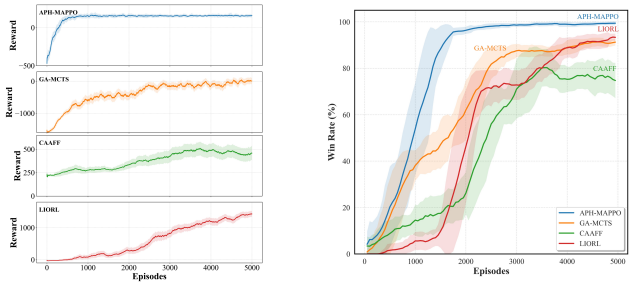


(a) Reward in the 3v3 scenario

(b) Win rate in the 3v3 scenario

Fig. 7. Reward and win rate comparisons in the 3v3 scenario

Fig 8 presents the corresponding comparison in the 5v5 scenario. Averaged over five random seeds, APH-MAPPO achieved the highest mean reward and win rate among the compared implementations, although the observed variability increased relative to the 3v3 setting. These results characterize the separately trained 5v5 configuration under the stated simulation constraints.



(a) Reward in the 5v5 scenario

(b) Win rate in the 5v5 scenario

Fig. 8. Reward and win rate comparisons in the 5v5 scenario

C. Component and Parameter Analysis

1. *Ablation Study*

To isolate the contribution of the principal components, we evaluated three 3v3 variants using the same architecture, optimizer, rollout length, episode limit, training budget, seeds, and evaluation protocol as APH-MAPPO. Specifically, APH was replaced with nearest-target assignment, the allied proximity and shared-exposure penalty was removed while retaining the other task and terminal reward terms, or valid-agent masking was disabled only during PPO updates.

Replacing APH with nearest-target assignment reduced the mean win rate by 52.50 percentage points, reduced the mean number of survivors, and increased shared-exposure risk. Removing the allied proximity and shared-exposure penalty reduced the mean win rate by 27.70 percentage points and increased its across-seed standard deviation from 4.14 to 28.68 percentage points, indicating lower and more variable cooperative performance under the evaluated 3v3 protocol. Disabling PPO masking produced a smaller 6.30-point reduction in mean win rate and a modest increase in variability. Because all variants exhibited similarly low proximity risk at the reported precision, the results do not support a strong component-level conclusion for that metric.

2. *Sensitivity Analysis*

A one-factor-at-a-time analysis examined the local effect of η, e_1, and e_2 in the 3v3 scenario. Starting from $(\eta, e_1, e_2) = (5, 20, 25)$, one coefficient was varied while the other two were fixed. Each configuration used the same single training seed and training budget, and its final checkpoint was evaluated over 200 episodes, as summarized in Tab V and Fig 9.

TABLE V

One-factor sensitivity analysis in the 3v3 scenario.

Coefficient	Lower value	Baseline	Higher value
η	3 : 91.5%	5 : 93.0%	7 : 91.0%
e_1	10 : 86.5%	20 : 93.0%	30 : 92.5%
e_2	15 : 88.5%	25 : 93.0%	35 : 89.5%

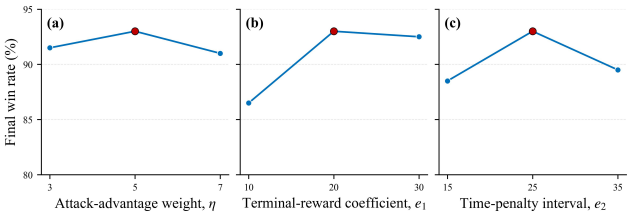


Fig. 9. One-factor sensitivity of the final win rate in the 3v3 scenario

The tested win rates ranged from 86.5% to 93.0%. Changes to η produced the smallest deviations from the baseline, whereas decreasing e_1 to 10 produced the largest reduction. The baseline setting achieved the highest observed win rate among the tested values. Because this sweep used one training seed, it is a local diagnostic rather than evidence of global optimality or multi-seed robustness.

D. Training Behavior

1. *Learning Curves*

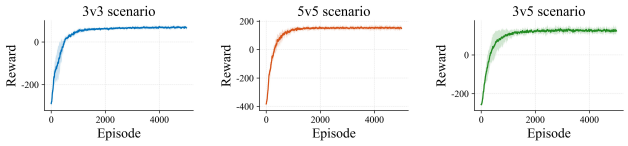
Fig 10 reports the mean episodic reward and one standard deviation over five independent training seeds.

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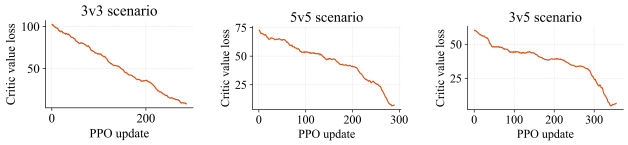
TABLE IV
Component ablation results in the 3v3 scenario (mean ± std. over five seeds). Win rate and survivors are higher-is-better; risk metrics are lower-is-better.

Method	Win (%)	Survivors	Proximity risk	Shared-exp. risk
APH-MAPPO	**88.10 ± 4.14**	**2.39 ± 0.35**	0.01 ± 0.01	**1.87 ± 1.11**
w/o APH	35.60 ± 12.49	1.74 ± 0.20	0.01 ± 0.00	4.91 ± 2.18
w/o allied penalty	60.40 ± 28.68	2.34 ± 0.34	0.01 ± 0.01	2.69 ± 1.13
w/o PPO masking	81.80 ± 6.00	2.38 ± 0.28	0.02 ± 0.02	2.27 ± 1.36



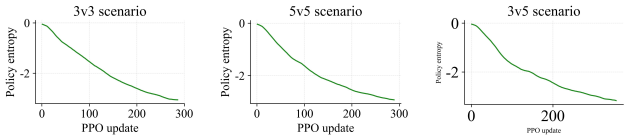
(a) 3v3 scenario (b) 5v5 scenario (c) 3v5 scenario

Fig. 10. Mean episodic reward across five training seeds. Shaded regions denote one standard deviation.



(a) 3v3 scenario (b) 5v5 scenario (c) 3v5 scenario

Fig. 11. Critic value-loss evolution during APH-MAPPO training.



(a) 3v3 scenario (b) 5v5 scenario (c) 3v5 scenario

Fig. 12. Policy-entropy evolution during APH-MAPPO training.

The curves improve during training and then remain within bounded ranges in all three configurations; the asymmetric 3v5 setting exhibits the lowest late-stage reward, consistent with its more difficult engagement geometry.

2. *Critic Loss and Policy Entropy*

To examine optimization behavior, we recorded critic value loss and policy entropy during diagnostic training runs. Fig 11 and 12 show 20-update moving averages.

The mean critic loss over the first versus last 20 logged updates decreased from 97.94 to 9.49 in 3v3, from 68.70 to 8.11 in 5v5, and from 58.05 to 5.80 in 3v5, after which it remained bounded. Policy entropy gradually decreased and stabilized near -3. Because the continuous policy is parameterized by Beta distributions, this quantity is differential entropy and may be negative as the action distribution becomes concentrated. These observations are consistent with lower critic-error magnitudes and a more concentrated action distribution over the logged training runs. They are diagnostic training trends, not evidence of theoretical convergence or direct measures of combat effectiveness.

E. Representative Engagement Cases

The following cases visualize individual evaluation episodes under the initial states listed in the corresponding tables. The trajectory panels show the spatial evolution, whereas the assignment-event panels report the locked-target changes and neutralization events. These cases are qualitative illustrations only and complement, rather than replace, the multi-seed aggregate results reported above.

1. *Three-on-Three Engagements*

a: *Random Initial Configuration*

Tab VI lists a representative randomized initial state, and Fig. 13 shows the resulting engagement.

TABLE VI
Initial state for the representative random 3v3 engagement.

Agent	x	y	z	v	ψ	μ	γ
UAV_0	5000	4500	5800	210	1.20	0.50	0.15
UAV_1	5000	6000	6200	205	-0.80	-0.30	-0.20
UAV_2	5000	7500	5500	215	0.40	0.80	0.10
Target_0	10000	4300	7000	185	-2.50	0.60	-0.25
Target_1	10000	5800	6500	190	1.80	-0.40	0.30
Target_2	10000	7800	7200	180	-0.30	0.90	-0.15

In the random case, assignments changed during the approach phase. UAV_2 neutralized Target 2 at step 67, and UAV_0 neutralized Targets 1 and 0 at steps 79 and 114; all friendly UAVs remained active.

b: *Head-On Initial Configuration*

Tab VII lists the head-on initial state, where the two teams approach each other directly.

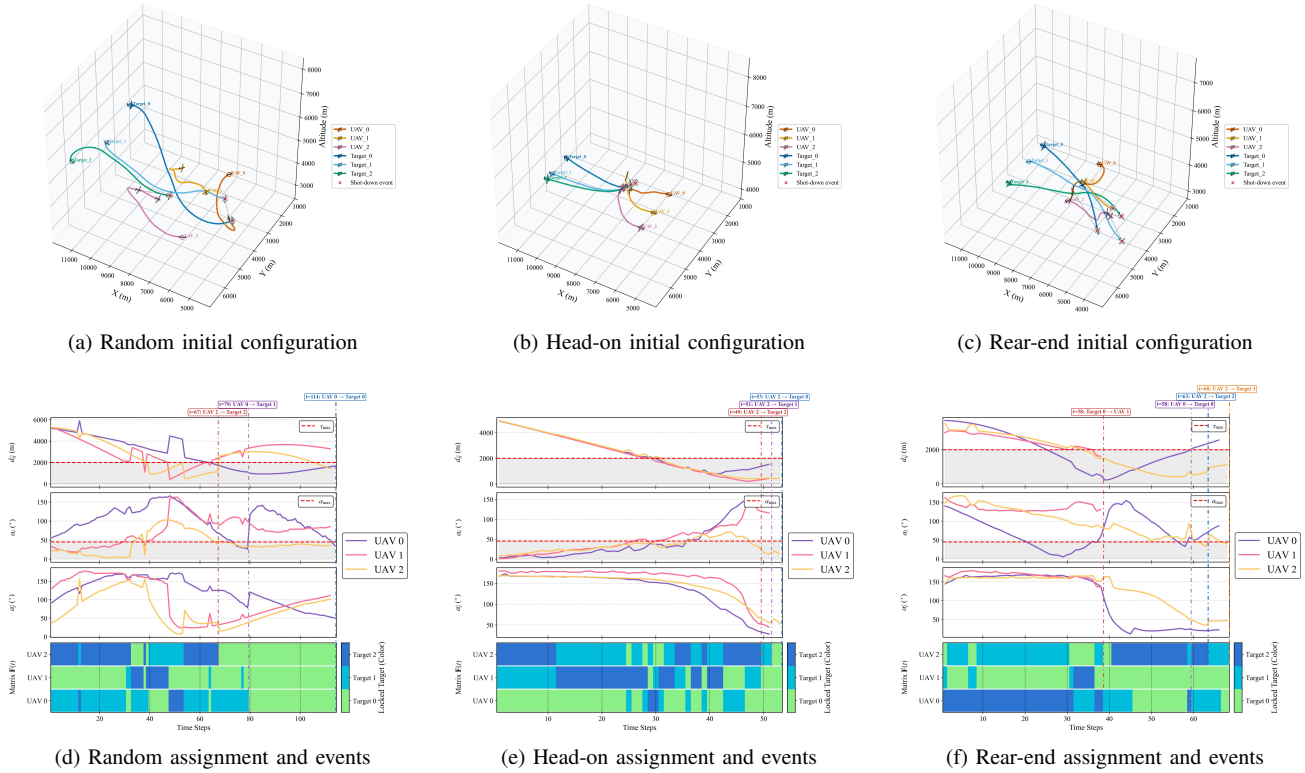
TABLE VII
Initial state for the representative head-on 3v3 engagement.

Agent	x	y	z	v	ψ	μ	γ
UAV_0	5000	4900	6000	200	0.01	0.87	0
UAV_1	5000	5900	6000	200	0.00	0.34	0
UAV_2	5000	6700	6000	200	-0.01	-0.32	0
Target_0	10000	5000	6000	200	3.04	-1.31	0
Target_1	10000	5900	6000	200	3.14	0	0
Target_2	10000	6200	6000	200	-3.14	1.31	0

In the head-on engagement, UAV_2 successively neutralized Targets 2, 1, and 0 at steps 49, 51, and 53. The

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(a) Random initial configuration (b) Head-on initial configuration (c) Rear-end initial configuration

(d) Random assignment and events (e) Head-on assignment and events (f) Rear-end assignment and events

Fig. 13. Representative 3v3 engagement trajectories and assignment events under three initial configurations.

assignment changed before the first neutralization as the initial symmetric geometry evolved.

c: *Rear-End Initial Configuration*

Tab VIII lists the rear-end initial state, where the friendly team starts ahead of the adversarial team.

TABLE VIII
Initial state for the representative rear-end 3v3 engagement.

Agent	x	y	z	v	ψ	μ	γ
UAV_0	5000	4900	6000	200	3.06	-1.20	0.25
UAV_1	5000	5900	6000	200	-3.12	0.60	-0.30
UAV_2	5000	6900	6000	200	3.10	-0.10	0.40
Target_0	8000	5000	6000	200	-3.04	0.80	-0.20
Target_1	8000	5900	6000	200	3.14	-0.50	0.35
Target_2	8000	7100	6000	200	3.07	0.30	-0.15

In the rear-end engagement, Target 0 neutralized UAV_1 at step 38. The remaining friendly UAVs completed the engagement, illustrating continued reassignment after a friendly loss.

2. *Five-on-Five Engagement*

The representative 5v5 initial state is listed in Tab IX, with trajectories shown in Fig. 14.

In the representative 5v5 episode, four targets were neutralized early and the remaining Target 0 at step 233. All friendly UAVs remained active. This qualitative trajectory is not statistical evidence of safety or effectiveness.

TABLE IX
Initial state for the representative 5v5 engagement.

Agent	x	y	z	v	ψ	μ	γ
UAV_0	4600	4000	5300	220	1.80	-0.80	0.20
UAV_1	5200	5200	6500	195	-0.90	0.60	-0.30
UAV_2	4800	6800	5800	230	0.40	0.00	0.00
UAV_3	5500	8200	7100	185	-1.60	-0.50	0.15
UAV_4	4900	9300	5000	205	2.20	0.40	-0.10
Target_0	9800	3800	7800	175	-2.20	1.10	0.20
Target_1	10500	5500	6200	210	1.40	-0.70	-0.15
Target_2	10000	7000	7500	190	-0.50	0.00	0.30
Target_3	10200	8600	5500	205	2.80	0.50	-0.10
Target_4	9500	9600	6800	180	-1.00	-1.00	0.10

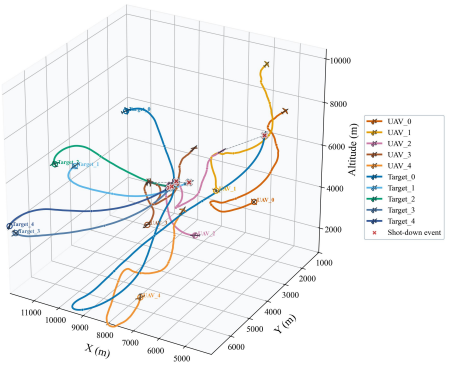


Fig. 14. Representative 5v5 engagement: 3D trajectory.

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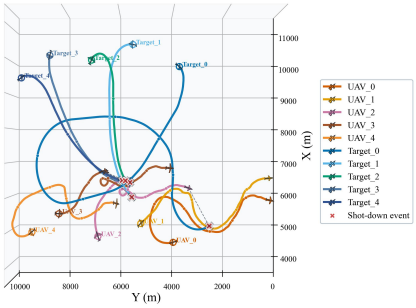


Fig. 15. Representative 5v5 engagement: planform trajectory.

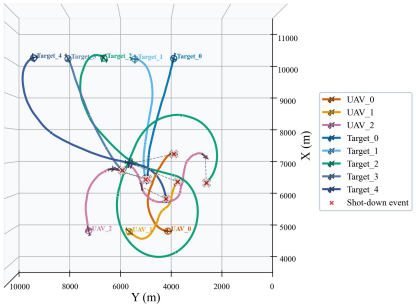


Fig. 17. Representative asymmetric 3v5 engagement: planform trajectory.

3. *Asymmetric Three-on-Five Engagement*

The representative 3v5 initial state is listed in Tab X, with trajectories shown in Fig. 16.

TABLE X
Initial state for the representative 3v5 engagement.

Agent	x	y	z	v	ψ	μ	γ
UAV_0	5000	4200	5800	210	1.50	1.20	0
UAV_1	5000	5600	6200	195	-2.20	-0.80	0
UAV_2	5000	7100	5500	220	0.30	0.50	0
Target_0	10000	4000	7000	185	2.80	-1.00	0
Target_1	10000	5400	6500	205	-1.60	0.60	0
Target_2	10000	6500	7200	190	0.90	-0.40	0
Target_3	10000	7800	6800	180	-2.50	0.80	0
Target_4	10000	9000	7400	200	1.80	-0.60	0

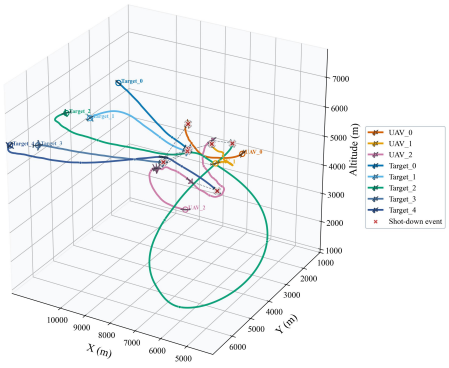


Fig. 16. Representative asymmetric 3v5 engagement: 3D trajectory.

In the representative 3v5 episode, UAV_2 neutralized Targets 0, 3, and 1 at steps 48, 50, and 52. After Targets 4 neutralized UAV_1 and UAV_0, UAV_2 neutralized the remaining targets and ended the episode as the sole surviving friendly UAV.

F. Evaluation Scope and Limitations

The reported results are restricted to the present simulation protocol. We adopt a fixed, rule-based, state-reactive stochastic-greedy adversary for three primary reasons: (i) it provides a controlled and reproducible baseline that enables us to isolate the performance of our proposed friendly-side cooperative algorithms; (ii) its 80/20 probabilistic action-selection rule—choosing the highest-scoring action with probability 0.8 and the second-highest with 0.2—mimics the decision logic of a realistic non-perfect opponent that is goal-directed but subject to occasional uncertainty; and (iii) it covers all 15 discrete action types in the maneuver library, ensuring that the friendly policies are exposed to a sufficiently diverse set of adversary behaviors during training. Accordingly, friendly policies were trained against this aggregate-sum 80/20 stochastic-greedy opponent, and the 3v3, 5v5, and 3v5 friendly policies were trained separately in a 3-DOF point-mass environment. During evaluation, the same fixed opponent policy is used without adaptation or learning.

We also conducted post-training sensitivity studies that varied either the action-selection probability or the rule used to aggregate pairwise opponent scores; however, all tested opponents remained hand-crafted and non-learning.

Consequently, these experiments do not establish zero-shot transfer across team sizes, robustness to independently learned or self-play opponents, or performance under higher-fidelity aerodynamic, sensing, communication, and weapon constraints. We explicitly acknowledge these limitations and identify the extension to learning-based adversaries as an important direction for future work.

V. Conclusion

In this paper, an APH-MAPPO framework is presented to address the multi-UAV cooperative air-combat problem. The framework combines attack-priority target assignment, residual Hungarian matching, and MAPPO maneuver control. In the specified 3-DOF simulation protocol, the selected 3v3 and 5v5 comparisons showed higher final mean reward and win rate for APH-MAPPO than the implemented baselines. The component ablations

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quantify the effects of the assignment and reward components under the evaluated settings.

Nevertheless, the considered confrontation scenarios remain simplified. The results are obtained against a fixed rule-based, state-reactive greedy opponent and do not establish robustness against learned or self-play adversaries. Moreover, the 3-DOF point-mass model omits aerodynamic, actuator, energy, and other related effects; therefore, the reported simulations do not demonstrate sim-to-real performance. Future work will consider a 6-DOF model and evaluate the framework against more diverse learned adversaries and in higher-fidelity hardware-in-the-loop environments.

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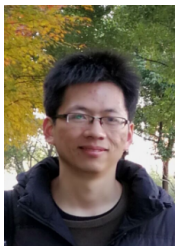
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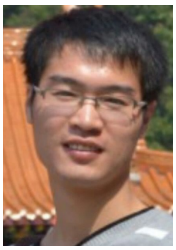
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